A Marker-based Method of Autonomous Landing for Unmanned Aerial Vehicles

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***Abstract*—In recent years, unmanned aerial vehicles (UAVs) have undergone rapid development and are widely utilized in fields such as rescue and goods delivery. Autonomous landing is a key problem for UAVs in these scenarios and this often relies on Global Navigation Satellite System (GNSS) or Real Time Kinematic (RTK). However, these systems require relatively expensive equipment and well conditions for GNSS signals. To achieve precise autonomous landing in the GNSS denied environment, a novel landing pad and pose estimation method based on ArUco markers are designed. We propose a new marker arrangement to facilitate accurate recognition at different distances, and also design an outlier removal method to integrate the results from multiple markers. We evaluate the localization accuracy of our method through various experiments and compare it with another vision-based method. The results indicate that our method improves the landing accuracy of UAVs while maintaining a low computational demand.**

Index Terms*—unmanned aerial vehicle, marker-based relative localization, robust recognition

I. Introduction

With the continuous expansion of UAV applications [1]–[4], the issue of UAV localization has received widespread attention. The landing phase is a critical stage in UAV missions and is also prone to errors. Researchers and engineers are therefore focusing on developing novel algorithms and technologies that can strengthen the UAV's ability to land accurately in different environments.

Currently there are two mainstream localization methods for UAV autonomous landing.The first type of landing methods is aided by global localization, such as GNSS, Light Detection And Ranging (LiDAR) SLAM and Visual Odometry (VO). GNSS based methods achieve centimeter-level accuracy but obtain poor performance in the areas with weak signals [5], [6]. The LiDAR-based approaches can accurately measure the distance from the UAV to obstacles, enabling precise estimation of its pose and ensuring accurate landing on the platform [7]–[9]. However, LiDAR sensors are relatively expensive and require substantial power resources. VO-based methods are cost-effective and don't rely on external signals [10]–[12]. However, they suffer from accumulated errors and require more computational resources. The second type is aided by relative localization, such as Ultra-Wideband (UWB) and fiducial markers. The UWB-based localization operates by transmitting short pulses of radio energy across a wide spectrum and enables the relative distance estimation. In [13], UAVs achieve accurate positioning and landing by supported ground robots and UWB radios, showing an impressive performance compared to Visual-Inertial Odometry. An energy-efficient uwb based method is proposed in [14], which supports running on a nano UAV and achieves precise localization. The UWB based methods usually have promising localization accuracy of centimeter-level, but the signal may be corrupted by multi path propagation and other interference. Fiducial markers are formed in high distinguishable patterns, which facilitate the detection and pose estimation even in challenging lighting conditions. Compared with global localization methods, the marker-based methods achieve short processing time and low computational burden, while maintaining comparative localization accuracy [15]. And compared to UWB-based methods, they avoid the significant power consumption and the negative impact brought by signal corruption.

Thus the marker-based methods have been widely investigated in recent years, including the new design of landing pad and the improvement of recognition algorithms. In [16], Salagame *et al.* designed an AprilTag landing pad to facilitate a wider range of detection and make the measurement easier.

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In [17], a landing pad consisting of eight ArUco markers surrounded by a circle is proposed, which ensures the detection in further range while the small marker embedded in the central marker ensures the visibility in close range. The above two studies are limited to the situation where the target board remains stationary, which restricts their widespread application. Reference [19] proposed an ArUco pad used for autonomous landing on the moving platforms. The vertical detection range of each marker is measured and the arrangement is well designed to facilitate accurate localization in dynamic scenarios. However, the results from different markers are simply fused by applying the same weights. In [20], the researchers designed an error model based on linear regression and weighted fusion is used to reduce localization error. However, the linear regression is sensitive to outliers and it is inevitable to generate outliers in dynamic scenarios.

In order to reduce the negative impact of outliers and improve the accuracy of UAV localization both when the landing pad is static and moving, we propose a novel marker-based method. The main contributions of our work are as follows:

- A landing pad organizing different sizes of markers in a new arrangement is proposed, which facilitates the robust detection in various heights and attitude angles.
- An outlier removal strategy based on Random Sample And Consensus (RANSAC) is designed. This eliminates the negative impact on localization from outliers and enhances the accuracy.
- Various experiments including an ablation study are conducted to show the effectiveness of our method both in static and dynamic scenarios.

II. The Design of Landing Pad

The landing pad is significant for UAVs as it provides a basis for recognition and pose estimation. In this paper, our landing pad is based on ArUco marker system [21]. We choose this system for its impressive performance in localization and low memory usage compared to other types of markers.

The overview of the landing pad is shown in Fig. 1. Our landing pad is consists of 61 ArUco 2D markers, including 4 large markers and 57 small markers. The different sizes of markers adapts to both the close-range field coverage and the long-distance recognition. The close-range detection is common in autonomous landing of UAVs. At the end of the UAV landing, due to the movement of the landing platform and the presence of control errors, any marker is prone to move out of the field of view, resulting in a loss of positioning information. This landing pad is designed with relatively small markers on all four sides to enhance the field coverage. At the same time, the central area of the pad is also designed with a cross-shaped distribution of small markers to continuously assist the UAV in locating the center point area of the target.

The long-distance recognition is essential for UAVs as they may have to reach a high altitude to perform the tasks and return back to the platform after mission completion. Therefore, the capability of long-distance recognition should be considered and the designed landing pad includes 4 large 2D markers to facilitate detection. With a dominant area of the landing pad, the large markers can simultaneously support robust recognition and long-distance positioning capabilities. In addition to the various types of markers used, their arrangement also plays a crucial role in the precise recognition of UAVs from different perspectives. Smaller markers are placed along the edges of the pad, optimizing the UAV's recognition when it views the pad from an oblique angle. Larger markers are placed near the center to enhance recognition quality when the UAV views the pad at right angles.

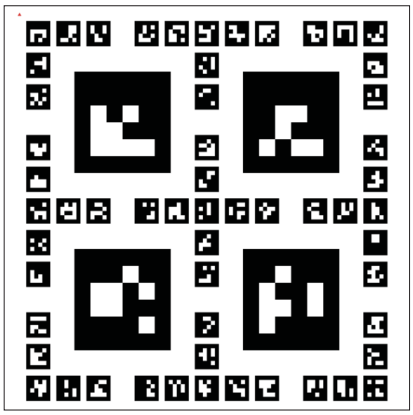


Fig. 1. The target pad based on ArUco markers.

III. The Relative Pose Estimation Algorithm Based on ArUco Target Pad

A marker-based relative pose estimation algorithm is introduced in this section, consisting of several parts: the image processing, the pose estimation of single marker and the fusion of pose results with a self-designed outlier removal strategy.

A. Image Processing

Image processing aims at extracting information from the captured images and sending it to the pose estimation module [22]. The processing procedure of ArUco markers mainly can be divided into three steps. The first step is to detect the candidate contours of ArUco markers, as the marker is enclosed by a square box. Due to the perspective transformation in images, the projected square on the image can form any quadrilateral but it must be convex. And we utilize this property for our detection. After candidate contours are filtered out, the corner points of them are stored for pose calculation. The second step is to decode the ArUco markers and obtain their ID numbers. The ArUco marker is composed of several black or white square blocks, each of which represents a single bit. Each specific arrangement of black and white squares corresponds to a unique ID number. The third step is to correct the corner points. The corner points detected in the previous steps are at the pixel level, which may introduce errors during subsequent pose estimation. Therefore, after detecting the corner points, it is necessary to refine them to obtain subpixel corner points.

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Then the coordinates of these corner points are sent to pose estimation module.

B. Relative Pose Estimation of A Single Marker

Pose estimation is the subsequent step in determining the relative pose of the downward camera based on the detected 3D corner points and their 2D projections, which is also described as a Perspective-n-Point problem in [23]. The coordinate systems used in the following discussions are defined in Fig. 2. In the world frame, the origin is set at the center of the ArUco board, and the origin of the inertial reference frame is set at the starting point of the UAV. Given the coordinate of a

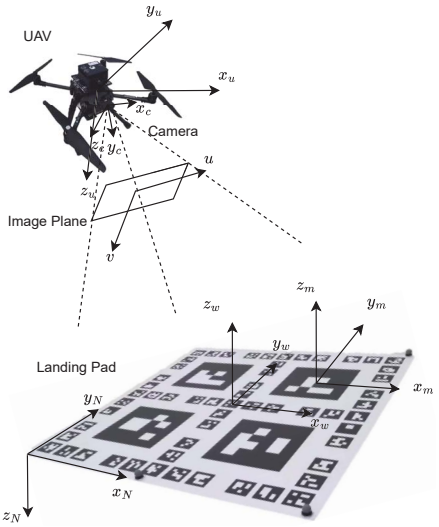


Fig. 2. The definition of coordinate systems, including the UAV body frame U, the camera frame C, the image frame I, the marker frame m, the world frame W and the inertial reference frame N.

corner point in the marker frame $\boldsymbol{P}_m = [X_m, Y_m, Z_m]^T$, the coordinate projected to the image plane $[u, v]^T$, the estimated depth s and the intrinsic matrix $\boldsymbol{K}$, the projection model can be described as:

$$s \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \boldsymbol{K}\boldsymbol{T}_m^c \begin{bmatrix} X_m \\ Y_m \\ Z_m \\ 1 \end{bmatrix}, \tag{1}$$

where $\boldsymbol{T}_m^c$ is the transformation matrix from the marker frame to the camera frame.

The model has a projection error because of unknown camera pose and noise. Then the camera pose is optimized by summing up and minimizing the error:

$$\boldsymbol{T}^* = \arg\min_{\boldsymbol{T}} \frac{1}{2} \sum_{i=1}^{n} \|\boldsymbol{u}_i - \frac{1}{s_i}\boldsymbol{K}\boldsymbol{T}\boldsymbol{P}_i\|^2. \tag{2}$$

Subsequently, the Levenberg-Marquardt algorithm is used to solve the constructed nonlinear optimization problem. The detailed process of gradient computation and optimization using Lie algebra is omitted here.

After solving the optimization problem, the transformation matrix $\boldsymbol{T}_m^c$ from a single marker frame to the camera frame is obtained. When more than one marker is detected, the estimated poses of these markers are fused to get a more accurate result. Before fusing them, the pose $\boldsymbol{T}_m^c$ should be transformed to the world frame. Given the transformation matrix $\boldsymbol{T}_w^m$ from world frame to the marker frame, the coordinates in the world frame $\boldsymbol{P}_w$ can be calculated using the following formula:

$$\boldsymbol{P}_m = \boldsymbol{T}_w^m \boldsymbol{P}_w = \begin{bmatrix} 1 & 0 & 0 & dx \\ 0 & 1 & 0 & dy \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X_c \\ Y_c \\ Z_c \\ 1 \end{bmatrix}, \tag{3}$$

where $\boldsymbol{P}_m$ are the coordinates in the marker frame, dx and dy refer to the x and y directional distances from the board center to the detected marker center, respectively. And the relative pose from the world frame to the camera frame can be obtained by:

$$\boldsymbol{T}_w^c = \boldsymbol{T}_m^c \begin{bmatrix} 1 & 0 & 0 & dx \\ 0 & 1 & 0 & dy \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \tag{4}$$

IV. Pose Fusion with an Outlier Removal Strategy

After the center transformation, the estimated poses from all the detected markers are fused. To remove potential outliers, we design a RANSAC-based fusion algorithm. The pseudo code is shown in Algorithm 1. In this fusion algorithm, the set of estimated poses S_{all} and other parameters are first initialized. Then we extracted the translational vector $\boldsymbol{t}$ and the yaw angle γ from the relative pose $\boldsymbol{T}_w^c$, and other components of the relative pose are omitted. We repeatedly sample and extract the subset S_{sub} of the estimated poses and get the weighted result $\boldsymbol{t}_{avg}$ and γ_{avg}. The weight w_i of the i-th estimated pose is defined by the detection scores of all poses:

$$w_i = \frac{s(p_i)}{\sum_{j=1}^{n} s(p_j)}. \tag{5}$$

The weighted result is compared with the last one and the error ϵ is calculated. The iteration is stopped when the error is less than the threshold ϵ_{th} or it reaches the maximum.

Algorithm 1 RANSAC-based pose fusion

Initialization: $S_{all}, S_{sub}, \epsilon_{th}, n_{max}, n \leftarrow 0, \epsilon \leftarrow 0, \gamma_{last} \leftarrow 0, \boldsymbol{t}_{last} \leftarrow (0,0,0)$
repeat
  $S_{sub} \leftarrow$ Sample(S_{all})
  for $i = 1$ to m **do**
    $\boldsymbol{t}_{avg} = w_i * \boldsymbol{t}_i$
    $\gamma_{avg} = w_i * \gamma_i$
  end for
  $\epsilon = Norm(\boldsymbol{t}_{avg} - \boldsymbol{t}_{last})$
  $\boldsymbol{t}_{last} = \boldsymbol{t}_{avg}$
  $\gamma_{last} = \gamma_{avg}$
  $n \leftarrow n + 1$
until $\epsilon < \epsilon_{th}$ **or** $n > n_{max}$
$\boldsymbol{t}_{res} = \boldsymbol{t}_{avg}$
$\gamma_{res} = \gamma_{avg}$
Output: $\boldsymbol{t}_{res}, \gamma_{res}$

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The fusion result is then transformed into the body frame of the UAV by left-multiplying the camera-body extrinsic matrix $\boldsymbol{K}_c^b$. Finally the global pose of the UAV $\boldsymbol{T}_N^u$ in the inertial reference frame is calculated as follows:

$$\boldsymbol{T}_N^u = \boldsymbol{T}_w^u * \boldsymbol{T}_N^w, \tag{6}$$

where $\boldsymbol{T}_N^w$ is the global pose of the target pad, $\boldsymbol{T}_w^u$ is the fusion pose from UAV body frame to the world frame.

V. Experiments and Discussions

In this section, two types of experiments are conducted to verify the accuracy of the proposed framework. We compare our target pad and pose estimation algorithm with other state-of-the-art frameworks, and then evaluate the localization accuracy in a laboratory equipped with a motion capture system.

A. Experiment Setup

The whole architecture of our experiment system is shown in Fig. 3.

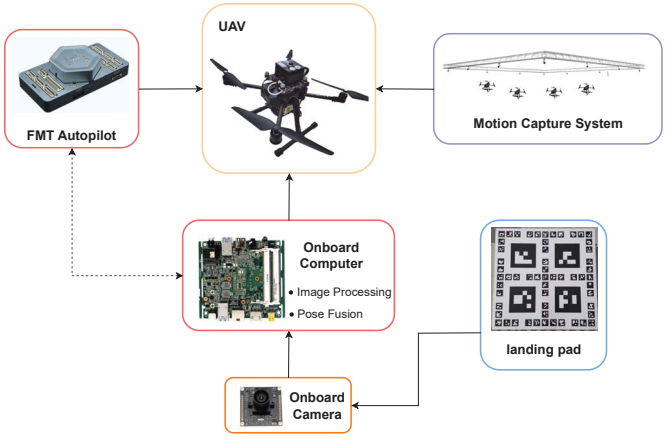


Fig. 3. The architecture of the landing experiment system.

We use a UAV for localization experiments and a motion capture system for providing the ground truth. The target pad is placed on a UGV for moving tests. The UAV is the AMOVLAB MFP450 quadrotor, which weighs 2200g and has a wheelbase of 410mm. The main sensor used in the experiments is a downward camera. The camera IMX290 is a 800 $\times$ 600 global shuttering color camera, with a FOV of 98° and its frequency is 30 Hz. All drivers and computations are running on an Intel NUC computer embedded in a UAV. Besides the sensor data and commands, the UAV receives the ground truth from the VICON motion capture system, which consists of 17 high-precision cameras and provides 6-DOF poses at a high frequency of 240Hz.

B. Accuracy Comparison of Different Marker Schemes

In this section, six types of ArUco [21], AprilTag [24] and ARTag [25] boards are tested to validate the accuracy and robustness of our configuration design and pose estimation algorithm, as shown in Fig. 4. In this evaluation, we designed three test cases: the hovering case, the deviation case and the takeoff-land case. In the hovering scenario, the UAV remains at a specific altitude of 0.6m while in the takeoff-landing scenario, the UAV ascends to a height of 0.6 m and then returns to the ground. The hovering case provides a relatively stable observation of the target boards while the takeoff-land case is more challenging with the varying detection range. In deviation case, the UAV deviates 0.4m from the center of the board, which is nearly outside of the board, resulting in significant changes in the detection view. In the evaluation of the other types of boards, we implemented the open-source detection and pose estimation algorithms from OpenCV. To test our target pad, we utilized the pose estimation algorithm proposed in this paper. Additionally, we compared the localization accuracy with and without the outlier removal algorithm.

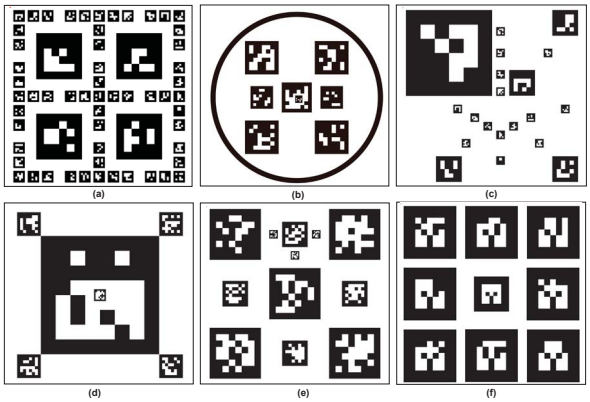


Fig. 4. Six types of target boards. (a) The ArUco board proposed in this paper. (b) The ArUco board proposed in [17] with a circle outer frame. (c) The ArUco board proposed in [26]. (d) The ArUco board proposed in [20]. (e) The AprilTag board proposed in [16]. (f) The ARTag board [25].

Table I illustrates the localization accuracy among the six target boards and algorithms. Noted that we conducted 5 independent trials and averaged the results to ensure the reliability of localization accuracy. In three cases, the UAV exhibits the smallest ATE errors using our board and pose estimation algorithm, achieving an accuracy of 0.0041m in the hovering case, 0.0083m in the takeoff-land case and 0.0203m in the deviation case. The results benefit from the excellent detection accuracy obtained by ArUco system and the specific configuration of the board. Our design allows the UAV to recognize the markers at a close range, which ensures the continuous detection during the landing process. The outlier removal strategy also enhances the performance by eliminating the impact of noise and misrecognition on localization. As seen in the table, the strategy enhances the localization accuracy especially when the UAV is moving, eliminating 18.62% of the error. Besides the localization accuracy, the average memory percent of ArUco marker is 10.30% while AprilTag uses 11.63% and ARTag uses 11.14%. This means that our scheme achieves high accuracy while maintaining lower memory usage.

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TABLE I
LOCALIZATION ACCURACY COMPARISON OF DIFFERENT TARGET PADS AND ALGORITHMS. (METRIC: RMSE).

Boards & algorithms	Hovering	Deviation	Takeoff-land
ARTag [25]	0.0148m	0.0881m	0.0245m
Niu *et al.* [26]	0.009m	0.0396m	0.0150m
Wang *et al.* [17]	0.0043m	0.0453m	0.0167m
He *et al.* [20]	0.0078m	0.0482m	0.0229m
AprilTag [16]	0.0081m	0.0243m	0.0114m
Ours w/o outlier removal	0.0046m	0.0206m	0.0102m
Ours	**0.0041**m	**0.0203**m	**0.0083**m

TABLE II
THE DATA PROCESSING STRATEGIES USED IN THE EXPERIMENTS.

Data Processing	Description
Time Alignment	The ground truth is selected as the time alignment benchmark, which means that the starting timestamp of other two types of data are aligned with our method.
Data Analysis	RMSE calculation is applied to analyze the averaged data which reflects the performance of the methods.
Data Visualization	To ensure that the three types of data are displayed on the same plot, interpolation is performed to align the higher frequency data with the lower frequency data.

C. Autonomous Landing Tests

This section presents the results of autonomous landing on the proposed landing pad in the vertical landing and following landing motion modes. Besides the ground truth obtained from the VICON motion capture system, we also compared our algorithm with another visual-based method VINS-Fusion. The legend gt in the figures represents the ground truth of the UAV, our represents the estimated values using ArUco markers, and vins denotes the pose estimation results obtained by VINS-Fusion. Table II shows the data processing strategies used in this experiment. Time alignment is utilized in the preprocess of data, as the start timestamp may be not align well in three different localization systems. During the data analysis, the RMSE calculation is applied to indicate the localization performance of our method and VINS-Fusion. And to display three types of data in the same plot, we resampled the high-frequency data to match the time intervals of the low-frequency data.

Fig. 5 shows the first experimental scenario when the UAV receives the command and begins to land from an altitude of 2.4m, keeping the landing pad stationary. The z-axis curve indicates the vertical movement of the UAV, while the x-axis and y-axis curves remain nearly horizontal and indicate a minor deviation from the center of the pad.

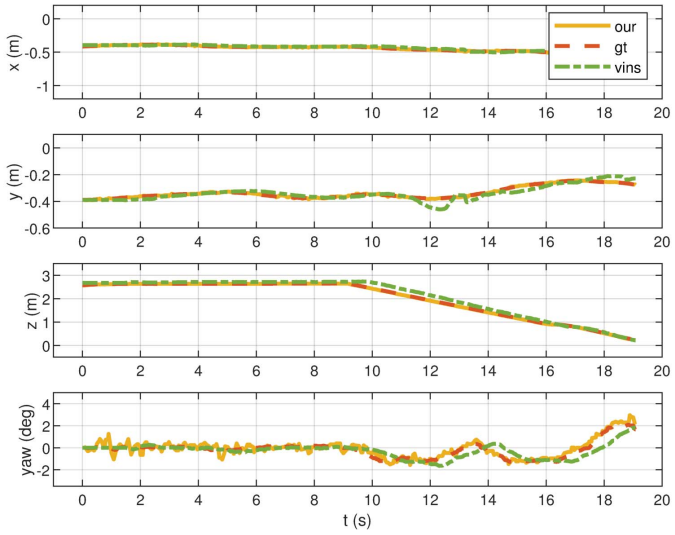


Fig. 5. Position and yaw angle estimation of the UAV during vertical landing.

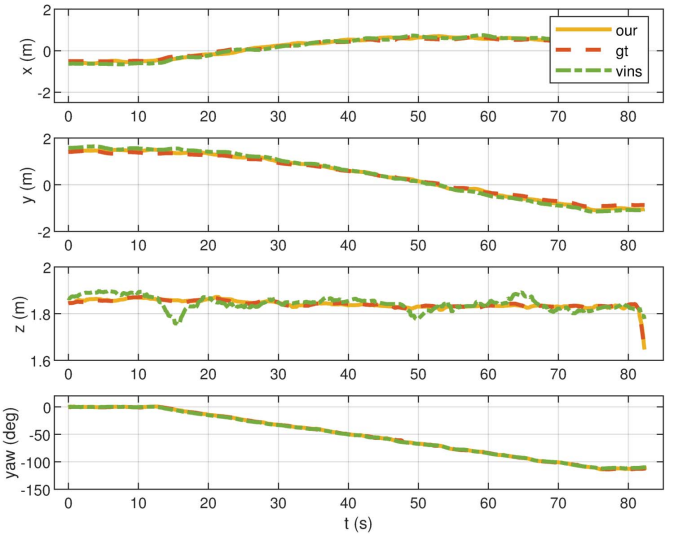


Fig. 6. Position and yaw angle estimation of the UAV during an follwing and landing motion.

TABLE III
THE COMPARISON OF DIFFERENT VISUAL-BASED METHODS IN TERMS OF POSITION AND YAW ERRORS. (METRIC: RMSE).

Scenes	VINS-Fusion		Ours	
vertical landing	0.0101m	0.4630°	**0.0079m**	**0.2511**°
following-landing	0.1720m	0.6737°	**0.1071m**	**0.4660**°

In Table III, the RMSE of the position error is 0.0079m, ensuring adequate localization accuracy to support the autonomous landing. Another visual-based method VINS-Fusion achieves an error of 0.0101m and this shows the better performance of our method when the UAV has minor deviations in x and y axes. Fig. 6 presents pose estimation during a successful motion that the UAV follows the UGV and lands back on the landing pad. The UAV maintains at a relatively low altitude to follow the UGV, and receives a landing command after 80s. It can be observed from the figure that the estimated y curve is similar to cosine waves, when the yaw curve descends from 0° to -180°. The shape of the curves indicate the UAV performs a semi-circular motion during the following phase. And another

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observation is that the altitude estimation of VINS-Fusion has slight oscillation, while our method correspond well to the ground truth. The estimation of VINS-Fusion on the y-axis also has a slight deviation from the ground truth, the above reasons lead to a relatively large RMSE value in its pose estimation.

In terms of landing accuracy, Our method achieves better performance from the above experiments, this partially benefits from the reasonable configuration of the markers. The small markers in the edge of the pad allow the UAV obtain precise poses even when it gets close to the pad, while the large markers help the UAV to localize itself even in high altitude. The wrong estimation values are inevitable during the dynamic scenarios but our ransac-based method selects the most reliable values to estimate the poses, eliminating the negative effect of outliers. Besides the landing accuracy, we also focus on the computational burden the methods may bring to the UAVs. With an approximate calculation, we have observed less than 10% CPU utilization when running our method on the NUC-12th computer, while visual localization methods like VINS-Fusion require up to 26% CPU utilization. This indicates that our method is more lightweight and saves more computational resources when used for UAV's autonomous landing.

VI. CONCLUSION

In this paper we present a marker-based method that consists of a novel target pad and a pose estimation algorithm with outlier removal. The method enhances the autonomous landing of UAVs both in static and dynamic scenarios, and it is much less sensitive to outliers. A series of experiments including the comparison with other marker schemes and visual odometry are conducted to show the robustness and accuracy of our method. In the future, the method will be expanded to more practical applications and can be further utilized in multi-robot collaborative localization.

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